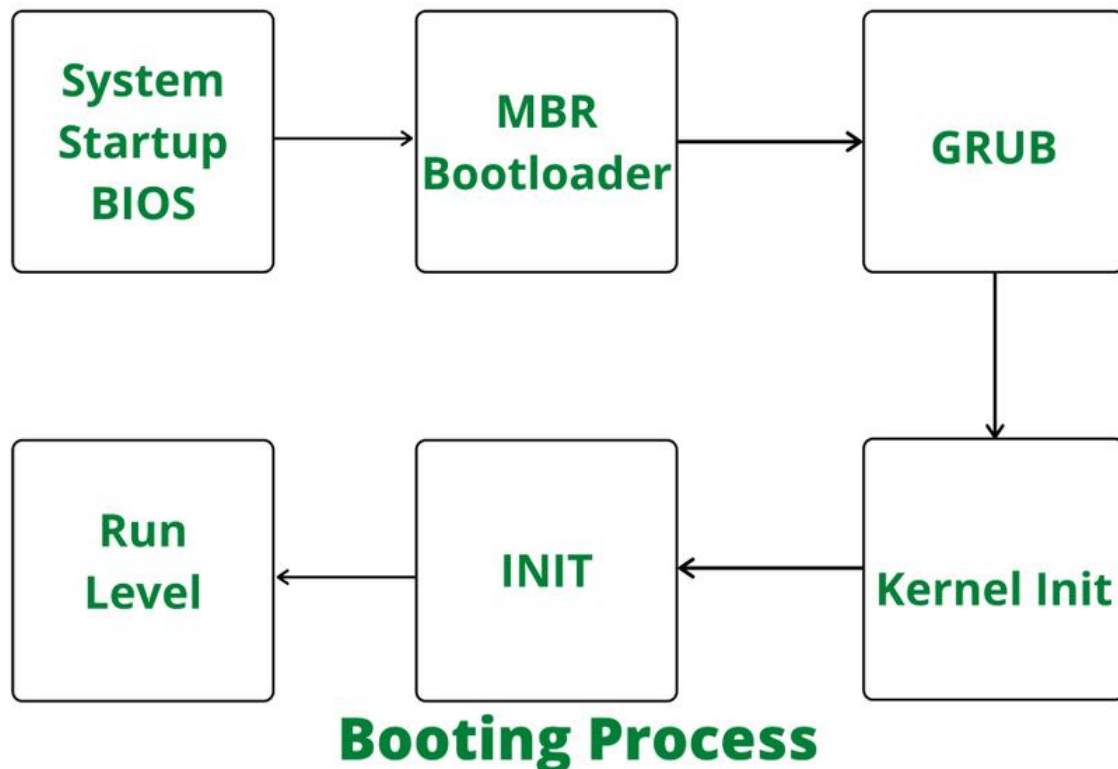


RHEL Boot Process-29

The **boot process** in Red Hat Enterprise Linux (RHEL) is the sequence of steps from powering on the system to reaching the login prompt. It involves both hardware and software initialization.

1. BIOS/UEFI Initialization

- When the system powers on, the **BIOS or UEFI firmware** runs.
- It performs **POST (Power-On Self Test)** to check hardware.
- It locates and loads the **boot loader** from the boot device (disk, CD, USB, or network).



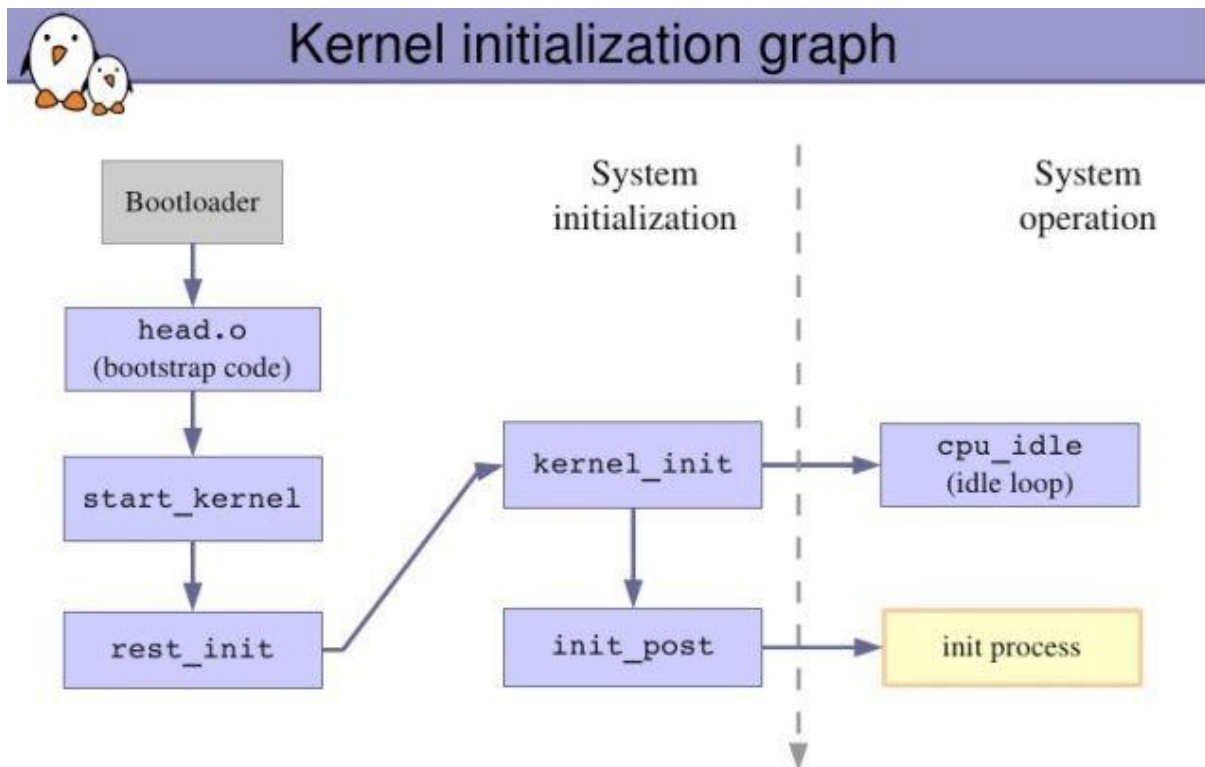
2. Boot Loader (GRUB2)

- **GRUB2 (Grand Unified Bootloader version 2)** is the default boot loader in RHEL.
- Responsibilities:
 - Provides a menu of available kernels.

- Loads the selected **Linux kernel** into memory.
- Passes control to the kernel.
- Configuration file: `/boot/grub2/grub.cfg`.

3. Kernel Initialization

- The **Linux kernel** is loaded into memory by GRUB2.
- Kernel initializes:
 - CPU, memory, and hardware drivers.
 - Mounts the **initial RAM disk (initramfs)** which contains temporary drivers for booting.
- Kernel mounts the real root filesystem.
- Then, kernel starts the **systemd process** (PID 1).



4. systemd Initialization

- **systemd** replaces the older SysV init.
- systemd is the first user-space process.

- Responsibilities:
 - Mount filesystems.
 - Start essential services.
 - Manage system targets (like runlevels).
- Default target in RHEL: graphical.target or multi-user.target.
- Configuration file: /etc/systemd/system/default.target.

5. Target and Services

- systemd brings the system to the specified **target**:
 - multi-user.target → command-line login.
 - graphical.target → GUI login.
- Loads unit files (.service, .target) to start services like networking, cron, etc.

6. Login Prompt

- After systemd finishes loading all services, you get:
 - **tty login prompt** (text mode).
 - or **graphical login manager (GDM)**.
- User enters credentials, shell is started, system is ready for use.